Virtual Cross Reference Matrix:

Requirement ID	Requirement Description	Verification Method	Verification Status	Evidence/Reference
FVS-1	Ingest Live Telemetry Data	Unit Testing	Completed	https://github.com//SME-3D-Flight-Visualization/tree/main/sme-3d-flight-visualization/test/unit/ingestion
FVS-2	Utilizes Modern Game Engine	Inspection	Completed	PDR Presentation
FVS-3	Render a 3D air vehicle position and orientation update in real time	Demonstration	Completed	CDR Presentation
FVS-4	Render and support for multiple air vehicle models	Demonstration	Completed	CDR Presentation
FVS-5	Maintain a smooth real-time experience, and must function at a minimum of 30 frames per second	Performance Testing	Completed	CDR Presentation
FVS-6	Provide the ability to record and replay data from a data stream to allow for post-flight review	Unit Testing	Completed	https://github.com/SME-3D-Flight-Visualization/tree/main/sme-3d-flight-visualization/test/unit/flightRecorder
FVS-7	Designed to run on the NVIDIA Jetson Orin Nano platform provided by the project sponsor	Demonstration	Completed	CDR Presentation
FVS-8	Support for multiple	Unit Testing	Completed	https://github.com/S

	camera views of the virtual environment (chase camera, fixed camera, and free camera)			ME-3D-Flight-Visualization/tree/main/sme-3d-flight-visualization/test/unit/cameras
FVS-9	Implement a GUI that allows users to start and stop data capture, select between air vehicle models, switch between playback modes, and control all configuration settings without modifying source code.	Unit Testing and Demonstration	Completed	https://github.com/SME-3D-Flight-Visualization/tree/main/sme-3d-flight-visualization/test/unit/GUI https://github.com/SME-3D-Flight-Visualization/tree/main/sme-3d-flight-visualization/test/unit/recordingPanel https://github.com/SME-3D-Flight-Visualization/tree/main/sme-3d-flight-visualization/test/unit/playback
FVS-10	Support the real-time display of telemetry variables with display variables defined through a configuration file.	Unit Testing and Demonstration	Completed	https://github.com/SME-3D-Flight-Visualization/tree/main/sme-3d-flight-visualization/scripts/GUI https://github.com/SME-3D-Flight-Visualization/tree/main/sme-3d-flight-visualization/scenes/GUI
FVS-11	The team shall provide documentation, including setup instructions, operating procedures, and a	Inspection	Completed	SBOM Document User Guide Document

	software bill of materials (SBOM)			
FVS-12	The application may use, but is not limited to, industry standards such as C++, C#, and Python coding languages.	Inspection	Completed	PDR Presentation
FVS-13	The project shall be designed using modular and open system methodologies.	Inspection	Completed	PDR Presentation
FVS-14	The team shall use GitHub or GitLab to encourage configuration management and collaboration.	Inspection	Completed	https://github.com/SME-3D-Flight-Visualization
FVS-15	The project shall be planned, designed, developed, and tested using Agile methodologies.	Inspection	Completed	PDR Presentation

Known Bugs and Fixes:

Bug Name:	Bug Description:	Patched Description:	Patched Progress:	Evidence/ Reference
Invalid Config From sender	Config from the sender is not the starting position of the drone		Incomplete	
Flight Path Visualization in playback	While in playback, the flight path renders when going back to	When a spike or jump is made in playback mode, it cancels the flight path and	Complete	Inspection of the flight path

	the old position, spikes and jumps instead of creating a smooth showcase of the path.	rebuilds the mesh to start at a new position.		
Plane Model Configuartion	Currently when switching between models after switching to the plane model cannot switch back to the drone model.			
The plane model switch during flight makes the model inaccurate	When switching vehicle models while in a flight session, the vehicle is displaced incorrectly, making it inaccurately show the model.			